OPTIMIZERS (How model learns?)

Goal: Update weights to reduce error

- **SGD** → Updates weights step by step using small data batches (simple but slow)
- **RMSprop** → Adjusts learning speed based on past gradients (faster than SGD)
- **Adam** → Combines momentum + adaptive learning (most commonly used)
- **AdamW** → Adam + better weight decay (reduces overfitting better)
- **Adadelta** → Automatically adjusts learning rate (no manual tuning needed)
- **Adagrad** → Gives more learning to rare features (good for sparse data)
- **Adamax** → Stable version of Adam for large gradients
- **Adafactor** → Memory-efficient Adam (used in large models like NLP)
- **Nadam** → Adam + Nesterov momentum (faster convergence)
- **Ftrl** → Good for large sparse datasets (used in ads/recommendation)
- **Lion** → Uses sign-based updates (new, efficient optimizer)
- **Lamb** → Scales learning for large batch training
- **Loss Scale Optimizer** → Prevents very small gradient issues (used in mixed precision)
- **Muon** → Experimental optimizer (less commonly used)